

PRODUCT REQUIREMENTS DOCUMENT

Lumina LMS — Multi-Tenant SaaS Learning Platform

Product Name	ARK University LMS	Document Date	April 29, 2026
Project Type	SaaS Web Application	Document Status	Final
Product Owner	Prit Ukey	Version	v1.0

1. OVERVIEW

- Product Name & Tagline:** ARK University LMS — "Train. Track. Transform."
- Type of Project:** Multi-Tenant SaaS Web Application
- Background / Context:** Organizations struggle to manage employee training efficiently across departments and roles using fragmented, outdated systems. Existing LMS platforms lack flexibility, multi-company support, and meaningful analytics — forcing HR teams to rely on manual tracking and disjointed tools.
- Goal / Objective:** Provide a scalable, modern LMS that enables organizations to manage courses, automate assessments, and track employee performance in real-time — reducing manual training overhead and improving learning outcomes across the board.

2. PROBLEM STATEMENT

- What problem exists today:** Corporate training is fragmented and inefficient. HR teams rely on spreadsheets, email chains, and legacy LMS tools that don't scale, lack real-time analytics, and require significant manual effort to maintain.
- Who is affected:** HR Managers, Company Admins, and Employees across all departments who require structured, trackable, and accessible learning workflows.
- Why current solutions fail:** Existing LMS platforms suffer from poor user experience, no true multi-tenant architecture, limited quiz customization, no bulk content upload support, and a lack of AI-enhanced features for content and quiz generation.

3. SOLUTION WE PROVIDE

- What we are building:** A scalable, multi-tenant SaaS Learning Management System with structured course management, an advanced quiz engine, real-time analytics dashboards, and AI-powered content tools.

- **How it solves the problem:** By centralizing course management under a single platform with role-based access, automating quiz grading and progress tracking, and supporting bulk content uploads via Excel/JSON — Lumina LMS eliminates manual effort and delivers measurable learning outcomes.
- **Key differentiators:** True multi-tenant SaaS architecture with full data isolation per company; Excel-based bulk quiz uploads; granular progress tracking with per-answer analytics; premium dark-mode UI with Stripe-like UX quality.

4. STAKEHOLDERS

- **Product Owner:** Prit Ukey
- **Developers:** Prit Ukey (Solo Full-Stack Developer)
- **Designers:** Prit Ukey (UI/UX Designer)
- **Business / Client:** Company Admins using the platform for employee training
- **End Users:** Employees, HR Managers, Super Admins

5. USER ROLES

- **Super Admin:** Platform-wide control — manages companies, billing, approvals, global settings, and onboarding flows.
- **Admin:** Company-level control — creates and manages courses, modules, quizzes, and all learning content within their organization.
- **HR Manager:** Assigns courses to employees, monitors individual and team progress, generates performance reports.
- **Employee / User:** Consumes learning content (videos, notes, assignments), attempts quizzes, views personal progress and certifications.

6. REQUIREMENTS

Functional Requirements

- **Authentication:** Secure JWT-based user authentication with role-based access control (Super Admin, Admin, HR, Employee).
- **Course & Module Management:** Full CRUD for Courses → Modules → Content (Videos, Notes, Quizzes, Assignments) with modular, extensible structure.
- **Advanced Quiz Engine:** Supports MCQ, Fill-in-the-Blanks, Short Answer, and Coding question types with automated grading and percentage-based pass/fail logic.
- **Bulk Quiz Upload:** Import quizzes in bulk via Excel / JSON format with strong server-side validation and error reporting.
- **Progress Tracking:** Granular tracking of video watch time, notes viewed, quiz attempts, and module completion with automatic status updates.
- **Analytics Dashboards:** Role-specific dashboards — Admin (platform overview), HR (employee progress), Employee (my learning journey).
- **Course Certification:** Automated certification issued only after full module completion is verified by the system.
- **Multi-Tenant Engine:** Full data isolation per company using company_id scoping across all models. Company onboarding with simulated payment flow.

Non-Functional Requirements

- **Performance:** Page load times and API response times maintained under 2 seconds under normal load.
- **Security:** JWT-based authentication, role-based access control, and scoped data access to prevent cross-tenant data leakage.
- **Scalability:** Multi-tenant PostgreSQL architecture with optimized queries and indexing to support growth across multiple companies and thousands of users.

7. PHASES

Phase	Focus	Key Deliverables	Duration
Phase 1: Core LMS (MVP)	Authentication, core course structure, and basic quiz functionality.	• JWT Authentication & RBAC • Course & Module Management • Basic MCQ Quiz System • Employee Progress View	Day 1 – Day 3
Phase 2: Advanced Features	Bulk uploads, analytics dashboards, and extended quiz types.	• Excel/JSON Bulk Quiz Upload • Real-time Analytics Dashboard • Fill-in-Blank, Short Answer & Coding Questions • Quiz Review UI (Your Answer vs Correct Answer)	Day 4 – Day 5
Phase 3: SaaS Scale	Multi-tenant SaaS architecture and platform polish.	• Multi-Company Onboarding & Billing • Course Certification System • Premium UI Redesign (Stripe-level UX) • Bug Fixes & Performance Optimization	Day 6 – Day 7

8. TECH STACK

- **Frontend:** Next.js 14+ (App Router), TypeScript, TailwindCSS, ShadCN UI, Framer Motion, Lucide React
- **Backend:** FastAPI (Python)
- **Database:** PostgreSQL with SQLAlchemy ORM
- **Validation & Auth:** Pydantic models, JWT-based authentication
- **Libraries:** Pandas (bulk upload processing), SQLAlchemy (ORM)
- **Hosting / Cloud:** Currently in active development — local environment (Localhost). Planned deployment: Vercel (Frontend), Cloud Server (Backend & DB).

9. TIMELINE & MILESTONES

Day	Milestone	Key Deliverables
Day 1	Project Foundation & Admin Panel	• Full folder structure setup (FastAPI backend + Next.js frontend) • Home page built and running • Basic Admin Panel with Login / Logout functionality
Day 2	Auth & Database Setup	• Dark theme UI implemented across the platform • JWT-based authentication integrated • PostgreSQL database connected and configured
Day 3	Multi-Role Panels & Core LMS	• HR and Employee user panels created • Role-specific dashboards built (Admin, HR, Employee) • Course and Module management system implemented
Day 4	Module Content Features	• YouTube lecture link integration inside modules • Notes feature added • Deliverables / Assignments system built • Quiz system created and linked to modules
Day 5	Dashboard Polish & Profile	• Digital / analytics-style dashboard improvements • Profile section built for users • Overall UI polished and refined with modern digital aesthetics
Day 6	Super Admin & Payment System	• Super Admin panel created with platform-wide controls • Multi-company onboarding flow implemented • Simulated payment system for company subscriptions
Day 7 (In Progress)	Excel Quiz Upload & Bug Fixes	• Bulk quiz upload via Excel sheet (eliminates manual quiz entry) • Bug fixes across modules and quiz system • Minor UI improvements and stabilization

10. RISKS & DEPENDENCIES

Risk / Dependency	Impact	Mitigation Strategy
Database Scaling Issues (Risk)	High: As companies and users grow, unoptimized queries can degrade performance significantly.	Implement optimized PostgreSQL queries, proper indexing, and connection pooling from the start.
Bulk Upload Errors (Risk)	Medium: Malformed Excel/JSON files could cause system crashes or silent data corruption.	Implement strict server-side validation with detailed error feedback and row-level error reporting.
UI Complexity (Risk)	Medium: Multi-role dashboards and advanced quiz UI increase development and maintenance complexity.	Follow iterative UX improvement cycles; use ShadCN component library to maintain consistency and reduce rework.

11. SUCCESS METRICS (KPIs)

- **User Adoption:** Achieve measurable Monthly Active User (MAU) growth across onboarded companies within the first 60 days.
- **Course Completion Rate:** Target $\geq 70\%$ course completion rate among assigned employees per company.
- **Quiz Performance:** Track quiz pass rates and average scores per course to measure content effectiveness.
- **System Performance:** Maintain API response times under 2 seconds under concurrent multi-company load.
- **Platform Engagement:** Track DAU/MAU ratio and average session duration as indicators of product stickiness.

12. PROJECT SIGN-OFF & AUTHORIZATIONS

By signing below, the stakeholders acknowledge that they have reviewed, understood, and agreed to the product specifications, phases, and requirements detailed within this Product Requirements Document.

PRODUCT OWNER

SIGNATURE

DATE

Name: Prit Ukey

Title: Product Manager

Project: Lumina LMS

DEVELOPER

SIGNATURE

DATE

Name: Prit Ukey

Title: Full Stack Developer

Project: Lumina LMS

TECHNICAL LEAD

SIGNATURE

DATE

Name: Prit Ukey

Title: Lead Engineer / CTO

Project: Lumina LMS

DESIGN LEAD

SIGNATURE

DATE

Name: Prit Ukey

Title: UI/UX Designer

Project: Lumina LMS